

F3 Deep Audit

Everything I Need to Know as an AI PM

Feature: F3 -- Policy Impact Mapping | Week 4 + Week 5 Day 1 (Days 22-29)

Eval accuracy: 73.3% (22/30) vs 80% CI gate target | Regression floor: 70% (held)

Audited: 2026-06-13 | RegWatch AI Build Session

Pipeline: policy section extractor -> dual-index embeddings -> hybrid BM25+dense matcher -> named-citation classifier
40% (Day 26) -> 73.3% (Day 27, named-regulation-match) -> 73.3% held (Day 29, one experiment kept, one rejected by regression CI)

This is a permanent reference document for F3 -- RegWatch's core/moat feature -- written by reading every line of every file. It captures what was built, why each layer of the matching/classification pipeline exists, how the eval framework works, and the complete 40% -> 73.3% accuracy journey, including the first real test of the Day 27 regression-CI gate on Day 29.

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2	Data Flow -- BSA-AML Policy & 4.2 traced end to end
3	The Matching & Classification Stack -- Days 24-29 evolution (UNIQUE TO F3)
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SECTION 1 -- Project File Map

7 F3 source files + golden eval set + 4 test files. For each: what it does, key class/function, key constants, connections, what breaks if deleted, and why this design over the obvious alternative.

FILE: src/f3_impact/extractor.py

DOES	Parses bank policy .txt files into PolicySection objects -- one per N.M numbered subsection, tagged with its parent SECTION N: header.
KEY FUNCTION/C LASS	PolicySection dataclass; extract_policy_sections(), extract_policy_file(), extract_policy_library()
KEY CONSTANT	SECTION_HEADER_RE, SUBSECTION_HEADER_RE
CONNECTS TO	Feeds build_indexes.py (policy_sections index)
BREAKS IF DELETED	Everything downstream -- no policy sections to embed, match, or classify
WHY THIS WAY	Regex parsing, not an LLM call. The 3 synthetic fixtures follow a consistent SECTION N: / N.M Title format; deterministic parsing is free, instant, unit-testable. LLM extraction is a documented fallback if real client policies don't follow this structure.

FILE: src/f3_impact/vectorstore.py

DOES	Local numpy+JSON vector store (VectorIndex) -- upsert_batch, query, save, load. Mirrors a Pinecone collection's interface.
KEY FUNCTION/C LASS	VectorIndex class
KEY CONSTANT	DEFAULT_EMBEDDING_MODEL (via f2_summarise.embeddings)
CONNECTS TO	Used by build_indexes.py and matcher.py
BREAKS IF DELETED	Both indexes (policy_sections, regulation_chunks) can't be built, saved, or queried
WHY THIS WAY	No PINECONE_API_KEY in .env. Reuses F2's local all-mpnet-base-v2 embeddings (zero cost, no data leaves the machine). Vectors pre-normalized so cosine similarity = a single dot product -- fast even for thousands of items on CPU. Swapping to real Pinecone later is a one-file change.

FILE: src/f3_impact/build_indexes.py

DOES	Builds + saves two VectorIndex collections to data/f3_indexes/: policy_sections (72 sections) and regulation_chunks (521 chunks from 25 summarised F1/F2 docs, via F2's chunk_hierarchical).
KEY FUNCTION/C LASS	build_policy_index(), build_regulation_index(), main()
KEY CONSTANT	INDEX_DIR, FIXTURES_DIR
CONNECTS TO	Reads extractor.py + F2's chunker.py + database.py; writes the indexes matcher.py reads
BREAKS IF DELETED	matcher.py has nothing to search
WHY THIS WAY	The 'dual-index' deliverable CLAUDE.md specifies -- one index per side of the match (policy text vs regulation text), each embedded/scaled independently. Day 29 added contextual retrieval to the policy-section embedding input (kept) and tried-but-rejected it for regulation chunks.

FILE: src/f3_impact/matcher.py

DOES	HybridMatcher -- for each policy section, finds top regulation chunks via dense search + BM25, combines with RRF, collapses chunk-level hits to one row per regulation document. build_matches() runs this for all 72 sections -> matches.json.
KEY FUNCTION/C LASS	HybridMatcher class (match_chunks, match_section); _rrf_combine(), _tokenize(), build_matches()
KEY CONSTANT	RRF_K=60, DENSE_TOP_K=20, BM25_TOP_K=20, CHUNK_TOP_K=15, MATCHES_PER_SECTION=5
CONNECTS TO	Reads both VectorIndexes; writes matches.json consumed by classifier.py
BREAKS IF DELETED	No candidate (policy section, regulation) pairs exist for the classifier to score
WHY THIS WAY	Hybrid (dense + BM25 + RRF), not dense-only -- same justification as F2 Day 16. Dense catches semantic matches; BM25 catches exact regulatory citations dense embeddings can miss. RRF_K=60 reuses F2's validated constant. dense_score kept separate from RRF score (which clusters near a ~0.03 floor) specifically for the classifier.

FILE: src/f3_impact/citations.py

DOES	Extracts the regulations a policy fixture EXPLICITLY CITES from its own text (Act / Regulation Letter / Abbreviation patterns). Caches per-policy results.
KEY FUNCTION/C LASS	extract_named_regulations(), get_named_regulations(), is_named_regulation_match()
KEY CONSTANT	_ACT_PATTERN, _REGULATION_LETTER_PATTERN, _ABBREVIATION_PATTERN, _cache
CONNECTS TO	Used by classifier.py and evals/f3_eval.py
BREAKS IF DELETED	classify_impact() loses its only signal beyond raw dense_score -- accuracy reverts to Day 26's 40%
WHY THIS WAY	Day 26's diagnosis: dense_score alone can't distinguish 'generic regulation sharing vocabulary' from 'a regulation this policy actually governs under'. Whether the policy itself names the regulation is a free, deterministic, auditable second feature -- no LLM call, no new training data. Limitation: only 3 citation styles, validated on 3 fixtures only.

FILE: src/f3_impact/classifier.py

DOES	classify_impact(dense_score, named_regulation_match) -> ImpactLevel (HIGH/MEDIUM/LOW/NOT_APPLICABLE). Adjusts dense_score by a named-match boost/penalty, then applies fixed thresholds. classify_matches() runs this over matches.json -> impact_results.json.
KEY FUNCTION/C LASS	ImpactLevel enum; classify_impact(), classify_matches(), main()
KEY CONSTANT	HIGH_THRESHOLD=0.55, MEDIUM_THRESHOLD=0.45, LOW_THRESHOLD=0.35, NAMED_MATCH_BOOST=+0.10, NO_MATCH_PENALTY=-0.20
CONNECTS TO	Reads matches.json + citations.py; writes impact_results.json (F3's actual output)
BREAKS IF DELETED	F3 has no output -- no impact levels for F4 to act on, nothing for Sarah's dashboard
WHY THIS WAY	Threshold rule, not a trained classifier (roadmap KM #17/#20 deferred -- no labeled data existed before Day 26). Thresholds are SR 11-7-auditable in one sentence: '0.47+0.10=0.57 >= 0.55 -> HIGH because Reg B is named in this policy's Regulatory Framework section.'

FILE: evals/f3_eval.py

DOES	Runs classify_impact() against 30 golden pairs, computes accuracy + confusion matrix, prints mismatches with rationale. Two gates: aspirational CI_GATE_THRESHOLD and measured REGRESSION_BASELINE.
KEY FUNCTION/C LASS	run_eval(), _load_dense_score_lookup(), _print_report(), main()
KEY CONSTANT	CI_GATE_THRESHOLD=0.80 (CLAUDE.md target), REGRESSION_BASELINE=0.70 (Day 27, measured floor)
CONNECTS TO	Reads fixtures/golden/impact_pairs.json + matches.json; calls classifier.py + citations.py
BREAKS IF DELETED	No automated check that F3 still works after a change -- eval-first build rule has nothing to enforce
WHY THIS WAY	Two separate thresholds on purpose: 80% = 'are we done' (currently red, expected, tracked). 70% = 'did we just break something that used to work' (CI-blocking). Day 29's Experiment B (70.0%, exactly at the floor) is the first real proof this distinction matters.

FILE: fixtures/golden/impact_pairs.json

DOES	30 hand-labeled (policy section, regulation) pairs with true_impact_level + one-line rationale, stratified across HIGH/MEDIUM/LOW/N-A and the Day 25 'generic-language over-match' failure pattern.
KEY FUNCTION/C LASS	n/a (data file)
KEY CONSTANT	_metadata.pair_count=30, _metadata.label_field='true_impact_level'
CONNECTS TO	Read by evals/f3_eval.py
BREAKS IF DELETED	The eval has nothing to score against -- run_eval() raises
WHY THIS WAY	CLAUDE.md names this exact path as F3's golden eval set. _metadata discloses labels are Claude-generated (v1), 'PENDING review by a compliance officer' -- not yet SME-validated ground truth (SR 11-7 caveat).

Test files (16 tests total, all passing)

File	Coverage
tests/test_f3_extractor.py (3 tests)	Synthetic sample parsing, multiline section bodies, all 3 real fixtures (72 sections, spot-checks
tests/test_f3_vectorstore.py (5 tests)	Upsert+query ranking, mismatched-length validation, empty-index query, __len__, save/load r
tests/test_f3_matcher.py (4 tests)	RRF combination logic, end-to-end match_section against a 3-doc fake index (best match wins
tests/test_f3_classifier.py (4 tests)	Threshold boundaries with/without named match (arithmetic spelled out in comments), classify
tests/test_f3_eval.py (4 tests)	Real golden set (total==30), controlled fake dataset (accuracy==0.5), CI_GATE_THRESHOLD

SECTION 2 -- Data Flow: BSA-AML Policy & §4.2 End to End

Tracing 'BSA-AML-Policy §4.2 -- Currency Transaction Reporting (CTR)' through the full F3 pipeline, from raw policy text to a HIGH-impact finding Sarah would see.

STEP 1: extract_policy_file()	Finds the line '4.2 Currency Transaction Reporting (CTR)' under 'SECTION 4: ...' -> PolicySection(policy_name='BSA-AML-Policy', section_id='4.2', section_title='Currency Transaction Reporting (CTR)', parent_section='SECTION 4: ...', text=<subsection body>).
STEP 2: build_policy_index()	Embeds (Day 29 contextual format): 'BSA-AML-Policy -- SECTION 4: ... Currency Transaction Reporting (CTR) <body>'. Vector stored under id 'BSA-AML-Policy::4.2'; metadata['text'] keeps the raw section text for BM25/evidence display.
STEP 3: HybridMatcher.match_section()	Dense search: cosine similarity vs 521 regulation_chunks vectors -> top 20. BM25: keyword overlap ('currency transaction report', '\$10,000', 'CTR') -> top 20. RRF combine (k=60). Collapses to per-document, keeping the best chunk as matched_chunk_text/matched_chunk_section_header and its cosine similarity as dense_score. Surfaces a FinCEN CTR-related document with dense_score ~ 0.6+.
STEP 4: is_named_regulation_match()	Checks whether BSA-AML-Policy's own text (its 'Regulatory Framework' section) names the Bank Secrecy Act / FinCEN regulations. It does -> named_regulation_match = True.
STEP 5: classify_impact()	$0.6 + \text{NAMED_MATCH_BOOST}(0.10) = 0.70 \geq \text{HIGH_THRESHOLD}(0.55)$ -> ImpactLevel.HIGH. Written to impact_results.json under §4.2's matches list with regulation_doc_id, regulation_title, dense_score, named_regulation_match, impact_level, matched_chunk_text.
STEP 6: evals/f3_eval.py	This (BSA-AML-Policy §4.2, FinCEN CTR doc) pair -- or one structurally identical -- is one of the 30 golden pairs labeled true_impact_level='high', and one of the 10/10 HIGH pairs the classifier gets right (Day 27 confusion matrix).
WHAT SARAH SEES	A HIGH-impact finding on 'BSA Policy §4.2 -- Currency Transaction Reporting (CTR)', citing the exact FinCEN regulation text that drove the match, with dense_score=0.70 and named_regulation_match=true as the auditable 'why' -- the Trust-Strategy §1/§3 pattern: show evidence, not just a verdict.

SECTION 3 -- The Matching & Classification Stack: Days 24-29

Unlike F2 (one retrieval stack, tuned once), F3's 'stack' is two coupled layers -- matching (which regulation chunks pair with a policy section?) and classification (given a match, how impactful is it?) -- built, broken, and fixed in sequence. Every measured change is below.

Day 24: Matching v1

HybridMatcher (dense + BM25 + RRF, RRF_K=60) built. 72 sections -> 252 raw matches.

RRF score clusters near floor (~0.03) -- not usable for classification directly. Flagged dev-DB coverage gap (25 docs lack CTR/SAR content for BSA).

Day 25: Matching v1.1 + Classification v1

Added dense_score (raw cosine similarity) alongside RRF score. First threshold pass: 27 high / 47 medium / 78 low / 99 N/A.

Found dense_score is a real signal (ECOA policy <-> 'Equal Credit Opportunity Act (Reg B)' correctly HIGH) but 'Reg B' also over-matches BSA/TRID sections -- flagged for Day 26.

Day 26: Eval v1 -- 40.0% (12/30), CI FAIL

Golden set (30 pairs) built and run for the first time. Root cause: generic, long documents over-match via generic regulatory vocabulary regardless of policy relevance.

Confusion matrix: HIGH 8/10 good; NOT_APPLICABLE only 2/11 (7 of 11 true negatives predicted HIGH/MEDIUM).

Day 27: Classification v2 -- 73.3% (22/30), CI FAIL but +33.3pts

citations.py (named_regulation_match) +
NAMED_MATCH_BOOST(+0.10)/NO_MATCH_PENALTY(-0.20) added. REGRESSION_BASELINE=0.70 added to eval.

HIGH 10/10, NOT_APPLICABLE 7/11. Remaining 8 mismatches all share one pattern: generic regulation vs unrelated policy, named_regulation_match=False, dense_score 0.45-0.61.

Day 28: Review day -- 73.3% unchanged

No matching/classification change. 10-pair MVP sample published with one deliberate known-error example.

Week 4 exit-gate scorecard: 2/6 met, 2 partial, 2 not met.

Day 29 Exp A (KEPT): Matching v1.2

build_policy_index() embedding input changed to '{policy_name} -- {parent_section}\n{section_title}\n{text}'.
73.3% (22/30) -- neutral, same confusion-matrix shape. Kept: no-cost, conceptually sound for larger future policy libraries.

Day 29 Exp B (REJECTED): Matching v1.3

`build_regulation_index()` embedding tried as 'Document: {title}\nSource: {agency}\nSection: {header}\n\n{chunk.text}'.

70.0% (21/30) -- fixed 2 of 8 mismatches (#9, #10) but broke 3 new ones (#11, #21, #30), net -1. Landed exactly on REGRESSION_BASELINE. NOT applied -- reverted, documented in `build_regulation_index()`'s docstring.

The Throughline

Every change after Day 25 was driven by a measured number, not intuition -- and Day 29 is the first time the regression-CI gate (built Day 27 specifically to catch exactly this) did its job on a real candidate change.

SECTION 4 -- Every AI/ML Decision

all-mpnet-base-v2 (reused from F2)	Same local sentence-transformers model for both policy sections and regulation chunks -- both sides must share an embedding space for cosine similarity to be meaningful. OpenAI/Anthropic embedding APIs would add cost + send (potentially client-confidential) policy text to a third party. Failure mode: if a real policy's vocabulary is far from this model's training distribution, dense_score quality degrades silently -- no drift monitoring yet.
Local VectorIndex (numpy+JSON), not real Pinecone	No PINECONE_API_KEY configured. Mirrors Pinecone's upsert/query interface so swapping later is a one-file change (same pattern as F2's embeddings, SQLite->Postgres). Failure mode: full matrix dot product on every query -- fine for 593 current vectors, won't scale to a real multi-tenant rollout (CLAUDE.md specifies Pinecone namespaces per client).
Hybrid search (dense + BM25 + RRF), not dense-only	Dense embeddings can miss exact regulatory citations ('12 CFR 1002.6', 'Regulation B') that differ only in punctuation/casing; BM25 catches these directly. BM25-only would miss semantic matches with different words/same meaning. RRF_K=60 reused from F2 Day 16. Failure mode: if BM25's tokenizer doesn't handle an unusual real-policy citation format, that match falls back to dense-only silently -- no alerting on BM25-zero-hit cases.
dense_score (raw cosine similarity) as classifier input, not RRF score	Day 24 found RRF score clusters near a ~0.03 floor across nearly all matches -- no usable dynamic range. dense_score has real spread (0.3-0.8+). Failure mode: dense_score is a SEMANTIC-OVERLAP signal, not a LEGAL-RELEVANCE signal -- Day 26 proved these diverge for generic regulations (high semantic overlap, zero legal relevance to an unrelated policy), which is exactly why named_regulation_match had to be added as a second feature.
Threshold classifier (classify_impact), not a trained model	No labeled training data existed before Day 26 (the golden set itself IS the first 30 labels) -- 30 examples is far too few to train without overfitting. Thresholds are SR 11-7-auditable in one sentence ('0.57 >= 0.55 -> HIGH because...'). Explicitly a v1 placeholder for roadmap KM #17/#20; any future trained classifier must preserve this audibility (e.g. via feature importances). Failure mode: a single global threshold/boost/penalty has a measured ceiling -- Day 27 found ~73-77% is close to the practical max for any single linear adjustment on this 30-pair set.

named_regulation_match via regex citation extraction, not an LLM call	Regex over each policy's own fixture text ('... Act', 'Regulation <Letter>', '(ABBR)'), cached per policy. An LLM-based check would add cost, latency, and a third non-deterministic AI decision per match -- harder to audit, and the task is genuinely pattern-matchable for current fixtures. Failure mode: only 3 citation styles. A real policy citing 'BSA requirements' without '(BSA)', or by CFR section number only, would get named_regulation_match=False for a regulation it clearly governs -- silently applying the wrong penalty.
Contextual retrieval (Day 29): policy-section context kept, regulation-chunk context reverted	Policy-section embeddings now include '{policy_name} -- {parent_section}\n{section_title}' as context (measured neutral, 73.3%, kept as a free bet for larger future policy libraries). Regulation-chunk context (document/source/section header) was tried and measured 70.0% -- fixed 2 known errors but broke 3 new ones, a net loss on this 30-pair set. Reverted. Failure mode: the 'kept' change is unproven on 30 pairs -- its value is an untested hypothesis pending more policies.

SECTION 5 -- The Classifier Calibration Journey

F2's 'Prompt Engineering Journey' tracked RAGAS scores across prompt versions. F3's equivalent is the classifier's adjustment-to-dense_score journey -- each step measured against the same 30-pair golden set.

v1 (Day 25) -- Pre-eval

Formula/change: `classify_impact(dense_score)` -- raw thresholds, no adjustment. Thresholds 0.55/0.45/0.35 chosen by inspecting the `dense_score` distribution of the 251 matches (27 high / 47 medium / 78 low / 99 N/A under this pre-eval split).

Measured impact: First pass -- thresholds picked by eyeballing score distributions, before any ground truth existed to validate against.

v2 (Day 26) -- 40.0% (12/30) FAIL

Formula/change: Same formula, now measured against the new 30-pair golden set. No code change.

Measured impact: This is the MEASUREMENT that revealed v1's thresholds badly over-predict impact for generic/long regulations matched against unrelated policies (NOT_APPLICABLE recall 2/11).

v3 (Day 27) -- 73.3% (22/30) FAIL, +33.3pts

Formula/change: `adjusted = dense_score + (NAMED_MATCH_BOOST if named_regulation_match else NO_MATCH_PENALTY)`, i.e. +0.10 or -0.20, then same 0.55/0.45/0.35 thresholds on adjusted.

Measured impact: The single biggest jump in F3's history. `named_regulation_match` gave the classifier a second, independent feature -- fixed all 3 true-HIGH false negatives (EOCA matches 0.47-0.52, just under threshold, now boosted ≥ 0.55) and 5 of 7 NOT_APPLICABLE false positives (generic-regulation matches now penalized below LOW).

v3.1 (Day 29, Exp A -- KEPT) -- 73.3% (22/30) -- neutral

Formula/change: Same v3 formula; `dense_score` itself shifts slightly because policy-section embeddings now include policy/section context.

Measured impact: Neutral -- same 22/30, same confusion-matrix shape. Confirms v3's thresholds are robust to small embedding-input changes on the policy side.

v3.2 (Day 29, Exp B -- REJECTED, not shipped) -- 70.0% (21/30) -- regression

Formula/change: Same v3 formula; `dense_score` shifts because regulation-chunk embeddings would include document/source/section context.

Measured impact: Regression. Fixed 2 of v3's 8 mismatches but broke 3 others -- net -1, exactly at REGRESSION_BASELINE. Reverted; documented as a rejected hypothesis in `build_regulation_index()`'s docstring.

The Remaining v3 Mismatch Pattern (8 pairs, unchanged since Day 27)

Every one involves 'Equal Credit Opportunity Act (Regulation B)' or 'Agency Information Collection Activities: Comment Request' matched against BSA-AML-Policy or TRID-Mortgage-Disclosure-Policy sections -- regulations that DON'T name a law those policies cite, but still score dense_score 0.45-0.61 on generic compliance-language overlap. The -0.20 penalty moves these into LOW where 6/8 should be NOT_APPLICABLE and 2/8 should be MEDIUM -- a spread no single linear adjustment can resolve (Day 27's 'ceiling' finding, re-confirmed by both of Day 29's experiments landing near this same set).

SECTION 6 -- The Eval Framework: How We Know F3 Works

Two Gates, Deliberately Different

CI_GATE_THRESHOLD = 0.80	REGRESSION_BASELINE = 0.70
CLAUDE.md's aspirational target	A MEASURED floor, set Day 27 at 73.3% minus a small margin
Currently FAIL (73.3%) -- expected to stay red until F3 clears it	Enforced by test_accuracy_does_not_regress_below_baseline on every test run
Answers: 'are we done?'	Answers: 'did we just break something that used to work?'
Tracked, not urgent while improving	CI-BLOCKING, independent of the 80% target

Key Insight, Validated Day 29

A number with no history can't be compared against; a number with history can. Experiment B (70.0%) technically passed the 70% gate but was a measured step backward from 73.3% -- only visible because 73.3% was on record from Day 27/28. The gate alone wouldn't have hard-failed; the HISTORY made the regression visible, and it was correctly not shipped.

Confusion Matrix as the Primary Diagnostic

Every run_eval() call prints a 4x4 true-vs-predicted matrix plus every mismatch with dense_score, named_regulation_match, true vs predicted level, and the golden set's rationale. This is what turned Day 26's '40%, something's wrong' into Day 27's specific, implementable fix.

Honest Caveat (carried through every layer)

The golden labels are Claude-generated (v1); _metadata in impact_pairs.json flags them 'PENDING review by a compliance officer'. 73.3% means '73.3% agreement with Claude's judgment of correctness' -- a meaningfully weaker claim than agreement with a compliance officer's judgment. Disclosed in docs/Trust-Strategy-v1.md as the first thing a design partner should help validate.

SECTION 7 -- Every Architectural Decision

Dual-index architecture (policy_sections + regulation_chunks as separate VectorIndex collections)	CLAUDE.md specifies dual-index Pinecone for F3. Separating the two lets each side be re-embedded/re-built independently -- Day 29's two experiments each touched only one index -- and keeps 'which side is this vector from' structural, not metadata-based.
matches.json and impact_results.json as separate artifacts from separate scripts	Matching (candidate generation) and classification (impact scoring) are conceptually different operations with different failure modes -- Day 24 found matching itself needed fixing (RRF score floor) before classification could be evaluated meaningfully. Separating them let Day 25's classifier work proceed on Day 24's matches without re-running the slower embedding/search step every time a threshold changed.
citations.py as its own module, not inlined in classifier.py	is_named_regulation_match() is reused by both classifier.py (to compute impact_level) and evals/f3_eval.py (to recompute the same feature for golden pairs, since matches.json may be stale relative to the golden set's dense_score_snapshot). A shared module guarantees both call sites use identical logic.
dense_score stored on every match, separately from RRF score	Without this (Day 24's original output), the classifier would have nothing usable to threshold. Storing both lets future work (e.g. a trained classifier, KM #17/#20) use either or both as features without re-running the matcher.
All F3 data artifacts (data/f3_indexes/*) gitignored and regenerable	Same pattern as F1/F2 -- build_indexes.py, matcher.py, classifier.py are each independently re-runnable (python -m src.f3_impact.<module>), so the repo stays small and the pipeline is reproducible from fixtures + the live database.
Fake-embedding-model test pattern reused across all 4 F3 test files	Avoids sentence-transformers model downloads in CI (same as F2's tests) while still exercising real ranking/threshold logic with predictable, hand-picked vectors.
REGRESSION_BASELINE as a second, separate constant from CI_GATE_THRESHOLD (Day 27, KM #258)	A single 80% gate would stay 'FAIL' indefinitely while F3 improves, providing no signal about regressions vs still-improving. Two thresholds let CI distinguish 'expected, tracked gap to target' from 'newly broken, blocking'.

SECTION 8 -- What Is Good, Weak, Missing

+ End-to-end pipeline runs on real data

72 real policy sections (3 synthetic fixtures) matched against 521 real chunks from 25 actually ingested and summarised F1/F2 regulatory documents -- not a toy/mock dataset.

+ Every output is evidence-backed

Every match carries `dense_score`, `named_regulation_match`, and `matched_chunk_text` -- Sarah never sees a bare 'HIGH' label (Trust-Strategy-v1.md §1).

+ Auditable classification

`classify_impact` is a deterministic function over documented constants -- same inputs always produce the same output, reasoning is one sentence (SR 11-7).

+ Regression CI works and has been proven

Day 29's Experiment B is a real, on-record example of a candidate change being measured, found to regress, and correctly not shipped.

+ 40% -> 73.3% in one day (Day 27)

From a single ~60-line module driven directly by the eval's confusion matrix -- the eval-first build rule did exactly its job.

+ Honest self-assessment built into product artifacts

F3-MVP-Sample-v1.md includes a deliberately-wrong example with explanation; the Week 4 exit-gate scorecard reports 2/6 met without softening.

! 73.3% vs the 80% CI gate

Not yet at CLAUDE.md target. The 8 remaining mismatches share one diagnosed pattern (generic regulation vs unrelated policy, `dense_score` 0.45-0.61); Day 27 concluded a single linear adjustment has hit its ceiling -- next step is a second feature or a trained classifier on the 30 labeled pairs (KM #17/#20).

! Golden set is Claude-labeled, not SME-validated

`_metadata` discloses this, but until a compliance officer reviews the 30 pairs, '73.3%' measures agreement with Claude's judgment, not ground truth. This is the first ask for any design partner (Trust-Strategy-v1.md).

! citations.py covers only 3 citation styles

Validated only against 3 synthetic fixtures. Needs testing against real client policy language before it can be trusted -- it's the single highest-leverage feature in the classifier.

! MEDIUM remains the hardest band

Only 3 of 30 golden pairs are MEDIUM, and the classifier gets only 1/3 right, unchanged since Day 26. Needs more MEDIUM examples in the golden set before this band's accuracy is even meaningfully measurable.

X No real policy upload UI

3 synthetic fixture policies only. Blocks: Week 4 exit-gate criterion 'upload 5+ policies'; any design partner with real policies can't use F3 yet.

X No F3 audit-trail logging

impact_results.json isn't logged with model/prompt version per CLAUDE.md's hard constraint, the way F2's AuditLog is. Blocks: should close before any design partner sees live output on real policies; prerequisite for F5.

X No feedback loop from Sarah back into the system

No path for 'Sarah marked this HIGH finding as wrong' to feed the golden set or thresholds. F4's HITL approval flow (Day 32) is a start but doesn't close this loop.

X Design partner outreach drafted but not sent

5 profiles + 2 email drafts exist (Design-Partner-Profiles-v1.md); per build rules, Claude drafts only. Blocks: real-world validation of whether 73.3% with 3 fixture policies is 'useful' to a real Sarah.

SECTION 9 -- 8 Interview-Ready PM Answers

Q1: Why is F3's accuracy 73.3% and not the 80% the CLAUDE.md target says?

73.3% (22/30) is up from 40% (12/30) one day earlier -- the named-regulation-match feature (Day 27) fixed most of Day 26's errors. The remaining 8 errors all share one root cause: regulations like 'Equal Credit Opportunity Act (Regulation B)' share generic compliance vocabulary with policies they don't actually govern, scoring `dense_score` 0.45-0.61 -- too high to be `NOT_APPLICABLE`, too ambiguous to cleanly separate from true MEDIUM cases with overlapping scores. Day 27's analysis concluded a single linear threshold adjustment has hit ~73-77% as its practical ceiling on this 30-pair set; closing the gap to 80% needs either a second independent feature or a trained classifier using the 30 labeled pairs that now exist.

Q2: What's the difference between the '80%' gate and the '70%' gate I see in the eval output?

80% (`CI_GATE_THRESHOLD`) is CLAUDE.md's TARGET -- currently failing and expected to keep failing until F3 actually reaches it; that's tracked, not urgent. 70% (`REGRESSION_BASELINE`) is a MEASURED FLOOR set the day we hit 73.3%, with a small margin -- it's CI-blocking. Any future change (different embedding model, threshold tweak, new fixture) that drops accuracy below 70% fails tests immediately, regardless of the 80% target. This distinguishes 'still working toward done' from 'we broke something that used to work'.

Q3: Day 29 says one experiment was 'kept' and one was 'rejected' -- what does that mean in practice, and did we lose a day's work?

No lost work -- both experiments are fully documented (notes/Day-29-F3.md and `build_indexes.py`'s docstrings) even though only one shipped. Experiment A (policy-name/parent-section context on policy-section embeddings) measured neutral -- 73.3%, same as before -- and was kept because it's free and conceptually sound for larger future policy libraries. Experiment B (document-title/source/section context on regulation-chunk embeddings) measured 70.0% -- it fixed 2 of the 8 known errors but broke 3 new ones, a net regression. It was NOT shipped. This is the regression-CI gate (built Day 27) working exactly as designed on its first real test case.

Q4: How does F3 decide a match is 'HIGH' impact, and can that be explained to a regulator?

Yes -- that's the design point. Every match has a `dense_score` (cosine similarity, 0-1) and a `named_regulation_match` boolean (does this policy's own text cite this regulation?). `classify_impact` adds +0.10 if `named_regulation_match` is true (or subtracts 0.20 if false), then compares to fixed thresholds (0.55/0.45/0.35 for HIGH/MEDIUM/LOW). The explanation for any output is one sentence: e.g. '`dense_score` 0.47 + 0.10 (Reg B is named in this policy's Regulatory Framework section) = 0.57 >= 0.55 -> HIGH.' No LLM is in this final step -- it's a deterministic function over two numbers and documented constants.

Q5: Why isn't F3 using a trained ML classifier yet, given the roadmap mentions one (KM #17/#20)?

Two reasons. First, there was no labeled data to train on until Day 26 built the 30-pair golden set -- and 30 examples is far too few to train a model without massive overfitting. Second, the threshold approach is explainable in a way that matters for SR 11-7: '`0.57 >= 0.55 -> HIGH`' doesn't need SHAP values or a black-box explanation. The threshold classifier is explicitly documented as a v1 placeholder; any future trained classifier must preserve this auditability property, not just match or beat 73.3% accuracy.

Q6: The Week 4 exit-gate scorecard says only 2 of 6 criteria are 'met' -- is F3 behind schedule?

Not behind in the sense that matters -- the two hardest technical bets (does the matching pipeline work on real data, and does the classifier produce explainable High/Med/Low/N-A output with section IDs) are both met. The 4 unmet/partial items are either scope deferred by design (5+ real policies needs an upload UI not yet on the roadmap), a number trending the right direction (40%->73.3% in one day, held for a good reason -- see Q3), or an action requiring the user, not Claude (sending design-partner emails).

Q7: What's the biggest single risk in F3 right now?

That citations.py's named-regulation-match feature -- currently the single highest-leverage signal in the classifier (it's the entire reason accuracy went 40%->73.3%) -- has only been validated against 3 synthetic fixture policies using 3 citation styles. If a real client policy cites regulations differently (e.g. by CFR section number only, or in a 'Related Regulations' table rather than prose), named_regulation_match would silently return False for regulations the policy clearly does govern, applying the -0.20 penalty incorrectly. This wouldn't error -- it would just quietly produce wrong impact levels.

Q8: If a design partner asks 'is this accurate,' what's the honest answer?

'73.3% agreement with Claude's own judgment on 30 example pairs that haven't yet been reviewed by a compliance officer -- and we can show you exactly which 8 pairs we get wrong and why.' That's the framing Trust-Strategy-v1.md and the Executive Deck's MVAP definition deliberately use instead of leading with the number: 'useful and honest today' rather than '80% accurate today.' The F3 MVP sample (F3-MVP-Sample-v1.md) practices this by including one of those 8 wrong examples right alongside the correct ones.

SECTION 10 -- Architecture Diagram

```

fixtures/policies/*.txt (3 synthetic policy fixtures: BSA-AML, Fair-Lending-ECOA, TRID)
|
v
extractor.py --> PolicySection[] (72 sections: N.M id, title, parent SECTION, text)
|
v
build_indexes.py --> build_policy_index()
|               | embeds "{policy_name} -- {parent_section}\n{section_title}\n{text}"
|               v
|               VectorIndex("policy_sections") [72 vectors]
|
+--> build_regulation_index()
|   | reads SUMMARISED RegulatoryDocuments (F1/F2, 25 docs)
|   | chunk_hierarchical() (F2) -> 521 chunks
|   | embeds raw chunk.text
|   v
|   VectorIndex("regulation_chunks") [521 vectors]
|
+-----+
|   matcher.py   |
| for each of 72 policy sections: |
| dense search (top 20) ---+      |
| BM25 search (top 20) ---+--> RRF (k=60)--+
|               |                  |
| collapse to per-doc, keep dense_score + |
| matched_chunk_text, top 5 docs/section |
+-----+
|
v
matches.json (~251 matches)
|
+-----+
|   citations.py   |
| is_named_regulation_match( |
| policy_name, regulation_title) | <-- regexes over
| (cached per policy) | fixtures/policies/*.txt
+-----+
|
v
+-----+
|   classifier.py   |
| adjusted = dense_score |
| + (NAMED_MATCH_BOOST if match) |
| + (NO_MATCH_PENALTY if no match) |
| classify vs 0.55/0.45/0.35 |
| -> HIGH/MEDIUM/LOW/NOT_APPLICABLE |
+-----+
|
v
impact_results.json
(24 high / 1 medium / 14 low / 212 N/A
across 72 sections, current build)
|
+-----+
v               v
docs/F3-MVP-Sample-v1.md   evals/f3_eval.py
(10-pair Sarah-facing sample) vs fixtures/golden/impact_pairs.json
                             (30 labeled pairs)
                             |
                             v
                             accuracy + confusion matrix

```

CI_GATE_THRESHOLD=0.80 (target, FAIL)
REGRESSION_BASELINE=0.70 (floor, enforced)

SECTION 11 -- The Eval Journey: 40% to 73.3%

The most important reference for future quality improvement. Documents exactly what was measured, what changed, and what the measured impact was at each step.

Day 26 -- Accuracy: 40.0% (12/30) [CI FAIL]

Golden set + eval pipeline built for the first time. This IS the baseline measurement -- no fix applied yet. HIGH 8/10 (80%) good; NOT_APPLICABLE only 2/11 (18%) -- 7 of 11 true negatives predicted HIGH/MEDIUM.

Day 27 -- Accuracy: 73.3% (22/30) [CI FAIL (target 80%), +33.3 pts]

citations.py + NAMED_MATCH_BOOST/NO_MATCH_PENALTY added to classify_impact. REGRESSION_BASELINE=0.70 added (KM #258). HIGH 10/10 (100%); NOT_APPLICABLE 7/11 (64%); LOW 4/6; MEDIUM 1/3 (unchanged).

Day 28 -- Accuracy: 73.3% (22/30) [Unchanged (Review day)]

No code change. 10-pair MVP sample + executive deck published; Week 4 exit-gate scorecard: 2/6 met, 2 partial, 2 not met.

Day 29 (Exp A, kept) -- Accuracy: 73.3% (22/30) [Unchanged]

Policy-section embeddings gain policy/parent-section context. Neutral but kept (low-cost, plausible future benefit).

Day 29 (Exp B, rejected) -- Accuracy: 70.0% (21/30) [At regression floor -- REVERTED]

Fixed #9, #10 (BSA §10.2, TRID §2.4 vs ECOA -- both now correctly NOT_APPLICABLE/LOW); broke #11, #21, #30 (new TRID-vs-ECOA mismatches). Regulation-chunk embeddings would gain document/source/section context. Net -1. Reverted, not shipped.

Day 27 Confusion Matrix (current production state, unchanged through Day 29)

true \ predicted	high	medium	low	not_applicable
high	10	0	0	0
medium	0	1	1	1
low	0	0	4	2
not_applicable	0	0	4	7

Day 26 Confusion Matrix (baseline being improved on)

true \ predicted	high	medium	low	not_applicable
high	8	2	0	0
medium	1	1	1	0

low	4	1	1	0
not_applicable	4	3	2	2

The trajectory in one sentence: 40% -> 73.3% in one day via a single ~60-line, zero-cost, fully-auditable feature; 73.3% held steady for 3 more days (1 review day, 2 experiments -- one neutral/kept, one regressive/rejected) -- the eval and regression-CI infrastructure built on Day 26/27 is now doing its job of keeping the number honest rather than just tracking it.

SUMMARY SCORECARD

Dimension	Score	Rationale
Engineering completeness	8/10	Full pipeline (extract -> dual-embed -> hybrid-match -> classify) runs end-to-end on real ingested F1/F2 data + 3 real policy fixtures, with 16 passing tests across 4 test files. Missing: real policy upload UI, scale beyond a local numpy vector store.
AI/ML quality	6/10	73.3% vs 80% target, with a clearly diagnosed and documented remaining error pattern (not a mystery). Every decision has a stated rationale AND a stated failure mode. Day 27's 'ceiling' finding (single linear adjustment maxes ~73-77%) is itself a valuable, honest result.
Eval rigor	9/10	Two-gate system (aspirational 80% + measured 70% regression floor) is a genuinely strong pattern, PROVEN on Day 29's real rejected experiment -- not just theoretical. Confusion matrix + per-mismatch rationale in every run. Only gap: golden labels are Claude-generated, not yet SME-reviewed (disclosed, not hidden).
Production readiness	4/10	No real policy upload, no F3-specific audit-trail logging (CLAUDE.md hard constraint not yet met for F3's outputs), local vector store not multi-tenant Pinecone, citations.py validated on only 3 fixtures/3 citation styles. All gaps documented, none silent.
PM explainability	9/10	Every classification decision reduces to one auditable sentence. The 40%->73.3% story, the Day 29 kept-vs-rejected experiments, and the honest Week 4 exit-gate scorecard are all genuinely good material for a design-partner or exec conversation -- arguably MORE compelling than a clean 80% would be.

Blockers for F4 (Task Generation)

- F4 will consume impact_results.json's HIGH/MEDIUM findings to generate tasks. Until F3's accuracy improves (or F4's HITL approval flow, Day 32, is in place), F4 would be generating tasks from a classifier that's wrong ~27% of the time on its own golden set -- manageable IF every F4 task requires human approval before action, but not if F4 assumes F3's output is ground truth.
- F3 has no audit-trail logging yet (model/prompt version + inputs per CLAUDE.md) -- F5 (audit trail) will need this retrofitted for F3's outputs, not just F2's.

Recommended Improvements Before Pilot

#	Improvement	Effort	Notes
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1	SME review of the 30 golden labels	Hard	Needs a compliance officer
2	Validate citations.py against citation styles beyond the 3 synthetic fixtures	Medium	1-2 days
3	Second classification feature or first trained-classifier pass (KM #17/#20)	Hard	Several days
4	F3-specific audit-trail logging, ahead of any design partner seeing live output	Medium	1 day